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Condition-based synthetic dataset for amodal segmentation of occluded cucumbers in agricultural images

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ABSTRACT

Occlusion, caused by overlapping leaves and dense foliage, poses significant challenges for crop segmentation in agricultural environments. Traditional segmentation methods struggle to handle these occlusions, particularly in complex and dynamic agricultural settings. As agricultural environments are subject to varying lighting conditions and environmental factors, the ability to detect and segment crops accurately remains a persistent issue in precision farming. This study aims to address these challenges through the development of a Synthetic Dataset Generation Framework that replicates realistic occlusion scenarios using a condition-based approach. Systematic adjustments to leaf composition, position, and occlusion ratios facilitated the framework to generate synthetic datasets representative of the diverse and complex conditions found in agricultural environments. To enhance realism, gamma correction-based brightness and contrast augmentations were applied to simulate both low-light and high-light conditions. These augmentations enhanced dataset diversity to better replicate real-world illumination variations. The ability of the model to handle dynamic lighting and complex occlusion scenarios was enhanced through this approach, advancing precision agriculture and contributing to sustainable farming practices by providing more reliable, adaptable segmentation solutions for real-world agricultural applications.

1. Introduction

Agriculture is undergoing a significant transformation with the integration of computer vision technologies (Sharma & Patil, 2018; Pandey et al., 2022). These advancements are increasingly being adopted to optimize key agricultural operations, such as sorting, harvesting, and crop monitoring. Computer vision technologies are reshaping modern agriculture by mitigating labor shortages, accommodating an aging workforce, and advancing sustainable practices (Tian et al., 2020; Ghazal et al., 2024). These systems facilitate precise crop identification and classification and provide the foundation for automation. They reduce reliance on manual labor while improving consistency, accuracy, and overall productivity (Zhao et al., 2016; Mavridou et al., 2019; Sivaranjani et al., 2022).

Recent advancements in deep learning have significantly propelled the development of computer vision, with convolutional neural networks (CNNs) and transformer-based architectures emerging as the two predominant approaches. These models demonstrate strong performance in agricultural applications, particularly for real-time detection, instance segmentation, and object recognition tasks (Tassis et al., 2021; Zhang et al., 2024; Sapkota et al., 2024). For example, YOLOv4 achieves accurate yield estimation by reliably detecting and counting ripe oranges under diverse lighting conditions (Mirhaji et al., 2021). Similarly, a modified YOLOv5 architecture facilitates efficient mobile-based tomato monitoring while maintaining optimal computational efficiency (Zeng et al., 2023). Performance enhancements have been achieved through architectural innovations, such as DenseNet-integrated YOLOv4, for improved cherry detection (Gai et al., 2023). For more complex scenarios, Mask R-CNN provides pixel-level segmentation capability, enabling precise analysis of overlapping or partially occluded crops (Xu et al., 2022; Hussain et al., 2023). These systems have demonstrated significant improvements in segmentation performance, with 97.31 % precision achieved in detecting overlapping apples (Jia et al., 2020) while simultaneously increasing strawberry harvesting efficiency (Pérez-Borrero et al., 2020). Specialized adaptations, including Strawberry R-CNN, have further advanced yield estimation accuracy through enhanced fruit counting capabilities (Li et al., 2023). These CNN-based models now serve as foundational tools for addressing key

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challenges in agricultural automation.

Transformer-based architectures—such as Swin Transformer and Vision Transformer (ViT)—have recently emerged as powerful tools for agricultural applications, demonstrating enhanced deep learning capabilities. Swin Transformer-based models exhibit superior performance in detecting small peaches within complex orchard environments (Seo et al., 2024). ViT models demonstrate significant potential for agricultural segmentation tasks through their ability to capture long-range spatial dependencies in image data; however, further optimization remains necessary for practical deployment in field conditions (Giménez-Gallego et al., 2024). Collectively, these technological advancements highlight the adaptability of deep learning in addressing diverse agricultural challenges—from precise fruit detection to comprehensive growth monitoring.

However, occlusion continues to pose one of the most persistent and technologically challenging problems in agricultural computer vision applications. In field conditions, target crops often become obscured by overlapping foliage, interwoven branches, or adjacent fruits, with visibility fluctuations influenced by growth stage, plant morphology, and cultivation practices (Paturkar et al., 2017; McCool et al., 2016; Yang et al., 2024). These occlusion challenges are further exacerbated by variable illumination conditions, complex leaf structures, and minimal color contrast between produce and surrounding vegetation.

Traditional segmentation methods—such as HSV-based thresholding, watershed algorithms, and manual feature extraction—often prove inadequate in complex agricultural conditions (Meyer, 1994; Yang et al., 2015). These approaches typically rely on assumptions of homogeneous backgrounds and distinct color delineations (Tamilarasi & Muthulakshmi, 2024), conditions seldom encountered in actual field environments with dense vegetation or variable shading (Niu et al., 2018). This limitation becomes particularly evident in cucumber detection, where minimal chromatic contrast between produce and foliage often leads to classification error and discontinuous segmentation masks (Guijarro et al., 2011; Barth et al., 2018).

Even CNN-based models, including Mask R-CNN and YOLO variants, face significant challenges when handling severe occlusion scenarios, primarily due to their dependence on visible features and limited receptive fields (Tian et al., 2019; Sood et al., 2021). Consequently, these models often demonstrate limited capability in reconstructing occluded regions, leading to compromised size estimation and unreliable object localization. These performance limitations validate a key finding from Akbar et al. (2024): deep learning models trained on limited-scale or open-field datasets often exhibit poor generalization when deployed in complex greenhouse environments, primarily due to challenges such as persistent occlusion, inconsistent illumination, and unique crop morphological features.

To address these limitations, researchers are increasingly investigating amodal segmentation approaches that predict complete object shapes despite partial occlusion (Li & Malik, 2016; Zhu et al., 2017). Agricultural adaptations demonstrate particularly promising outcomes: Gené-Mola et al. (2023) significantly enhanced apple size estimation accuracy through red–green–blue depth (RGB-D)-based amodal segmentation, while Yang et al. (2024) achieved 94.13 % mean intersection-over-union (mIoU) in tomato segmentation using a generative adversarial networks (GANs)-integrated transformer. Similarly, Chen et al. (2022) effectively reconstructed occluded citrus fruit boundaries using convolutional autoencoder networks.

However, existing amodal segmentation research has predominantly focused on radially symmetric fruits with uniform geometries. While recent studies have begun to explore non-circular crops such as cucumbers (Kim et al., 2023; Yin et al., 2025), robust methodologies capable of handling complex occlusion scenarios under systematic lighting variations remain underdeveloped for practical agricultural deployment. While Kim et al. (2023) developed a cucumber amodal segmentation method using synthetically generated occlusions, its moderate average occlusion rate (∼20 %) constrains practical robustness. Furthermore, prevailing methodologies often depend on artificially created datasets exhibiting limited diversity in illumination conditions, occlusion patterns, and plant morphologies (Gan et al., 2022; Sun et al., 2024), consequently reducing their field applicability.

Therefore, this study aims to develop a condition-based synthetic dataset generation framework for robust amodal segmentation of non-circular crops (particularly cucumbers) in complex occlusion scenarios. The proposed method systematically generates diverse occlusion patterns through controlled variations in leaf composition, spatial arrangement, and coverage density while incorporating adaptive brightness and contrast modifications to accurately simulate real-world illumination conditions in agricultural settings. This approach could enhance amodal segmentation model training, potentially facilitating more accurate inference of occluded cucumber regions and advancing the real-world applicability of vision-based systems in automated harvesting and precision agriculture.

2. Materials and methods

2.1. Framework overview

Fig. 1 illustrates the framework for generating synthetic occlusion datasets designed specifically for amodal segmentation tasks. The framework uses an annotated dataset containing cucumbers and leaves with bounding box labels as input, subsequently producing an occlusion-optimized training dataset for amodal segmentation models. The generation framework comprises two sequential processing stages: (1) the first stage, Data Preprocessing, involves executing multi-label annotation and individual object mask generation for all cucumber objects, and (2) Condition-Based Occlusion Mask Generation, where the system generates synthetic training images by strategically superimposing leaf occlusion masks onto cucumber targets while systematically varying three agriculturally-relevant conditions. This process produces a comprehensive training dataset encompassing diverse occlusion patterns, thereby enhancing the robustness and field generalizability of amodal segmentation models. Fig. 1 shows that the Data Preprocessing stage utilizes YOLOv10 and Segment Anything Model (SAM) v2 to generate multi-label annotated datasets with corresponding instance masks. Subsequently, the Condition-Based Occlusion Mask Generator applied three agriculturally relevant parameters—leaf composition, leaf position, and occlusion ratio adjustment—to simulate realistic occlusion scenarios. The resulting synthetic dataset accurately represented the variability and complexity of occlusions commonly encountered in cucumber-growing environments.

2.2. Data preprocessing for occlusion image synthesis

2.2.1. Data acquisition

For this study, 723 RGB images containing cucumbers and leaves were sourced from the publicly available AI-Hub platform. Each image included bounding box annotations for two object classes: cucumber and leaf. The initial dataset comprised 449 cucumber instances and 274 leaf instances, with each image containing a single-object annotation.

2.2.2. Dataset augmentation and segmentation mask generation

The preprocessing stage focused on increasing the number of cucumber and leaf instances in the dataset and generating segmentation masks. To achieve this, the YOLOv10 object detection model was first trained on the original dataset. Subsequently, the trained model was used to predict new cucumber and leaf instances to augment the dataset. After manual verification and annotation, the updated dataset contained 461 cucumber and 294 leaf instances (Table 1).

For mask generation, the SAM2 segmentation model was used to create pixel-level masks based on the bounding box annotations. These masks facilitated the accurate extraction of cucumber and leaf regions, which were then stored for the subsequent occlusion synthesis stage.

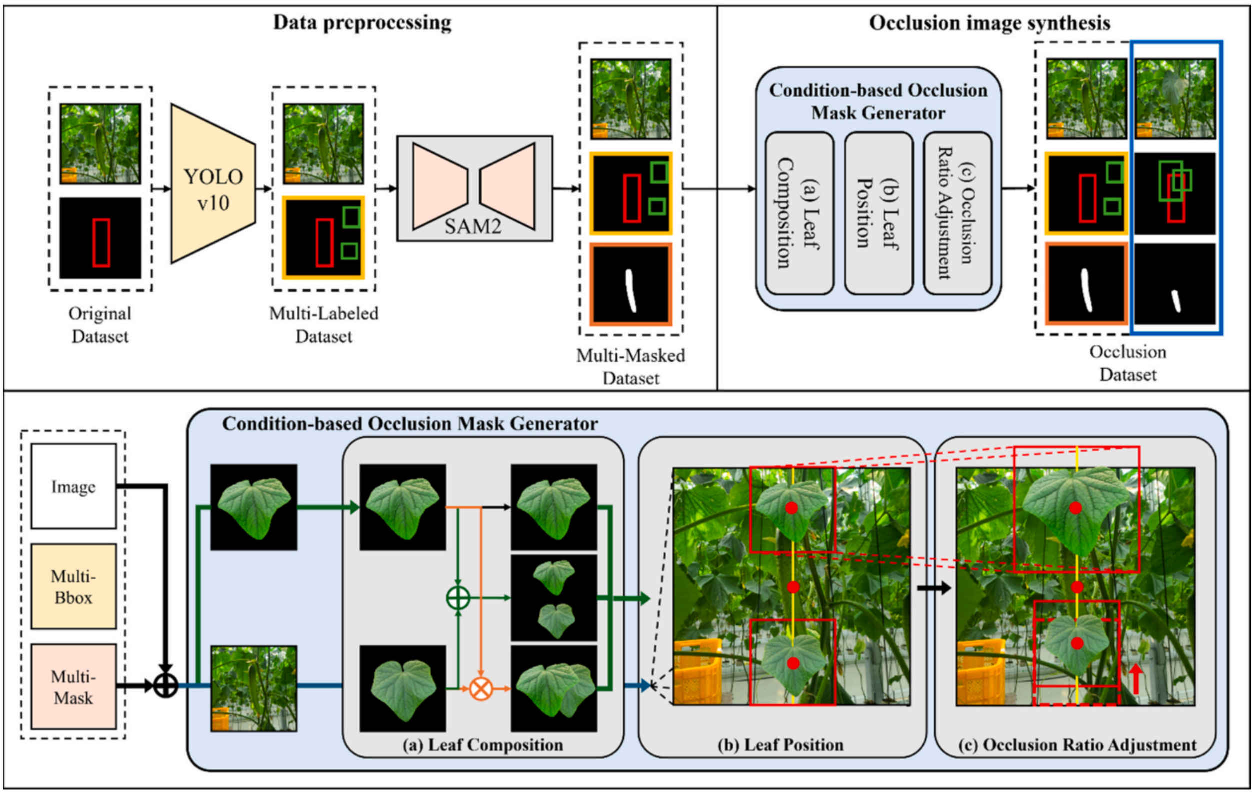


Fig. 1. Overview of the proposed framework for generating an occlusion dataset. The framework comprises two sequential stages—Data Preprocessing for multi-label and multi-mask dataset generation and Condition-Based Occlusion Mask Generation that creates realistic occlusion agricultural scenarios through systematic variation of leaf composition, spatial positioning, and occlusion ratios.

Table 1
Instances in the original, augmented, and synthetic datasets.

Dataset version	Cucumber instances	Leaf instances	Total instances
Original	449	274	723
Augmented	461	294	755
Synthetic	8,008	10,010	18,018

This process ensured the availability of clean, well-defined object masks, essential for generating realistic occlusion scenarios.

2.3. Synthesis of occlusion images

2.3.1. Generation of condition-based occlusion scenarios

The occlusion image synthesis process was specifically designed to replicate the complex and varied occlusion patterns found in actual cucumber cultivation environments. Based on previous studies of cucumber leaf morphology, growth stages, and canopy structure (Robbins & Pharr, 1987; Van Eck et al., 1998; Ding et al., 2020), we developed condition-based scenarios representing different levels of occlusion severity.

Table 2
Configuration settings for the condition-based occlusion mask generator.

Condition	(a) Leaf composition	(b) Leaf position	(c) Occlusion ratio adjustment
Baseline	Single leaf	Center	Fixed at 50 %
Condition (1)	Single leaf	Top or bottom	Increased leaf size
Condition (2)	Two separate leaves	Top and bottom	Position-based adjustment
Condition (3)	Two attached leaves	Top or bottom	Increased combined leaf size

Table 2 shows the configuration of the condition-based occlusion mask generator. The scenarios were organized into one baseline and three condition-based configurations. The baseline scenario involved placing a single leaf centrally over the cucumber with a fixed 50 % occlusion ratio, serving as a reference for comparison. Condition (1) involved randomly positioning a single, enlarged leaf at the top or bottom of the cucumber to simulate increased occlusion. Condition (2) featured two separate leaves at both the top and bottom of the cucumber, introducing dispersed occlusion from multiple directions. Condition (3) used two attached leaves positioned at the top or bottom of the cucumber, creating complex overlapping foliage and denser occlusion patterns.

Fig. 2 illustrates sample images and their corresponding modal masks for each scenario. This condition-based synthesis approach facilitated the generation of a robust and realistic training dataset, improving the adaptability of segmentation models to real-world agricultural conditions.

2.3.2. Construction of amodal segmentation datasets

The amodal segmentation dataset was constructed by systematically designing occlusion scenarios that balanced realistic field conditions and controlled experimental parameters. Leaves were strategically positioned to avoid occluding the cucumber centers in all conditions except the baseline, which purposely included central occlusion. This strategic placement enabled the model to learn from both typical and challenging occlusion patterns. Building on established research in cucumber leaf morphology and growth stages (Ding et al., 2020; Ota et al., 2007), we incorporated three key parameters: such as leaf size, occlusion ratio, and spatial arrangement. The occlusion ratios were implemented at 50 %, 75 %, and 90 % in a 5:4:1 proportional distribution to accurately represent different growth phase conditions throughout. Based on these

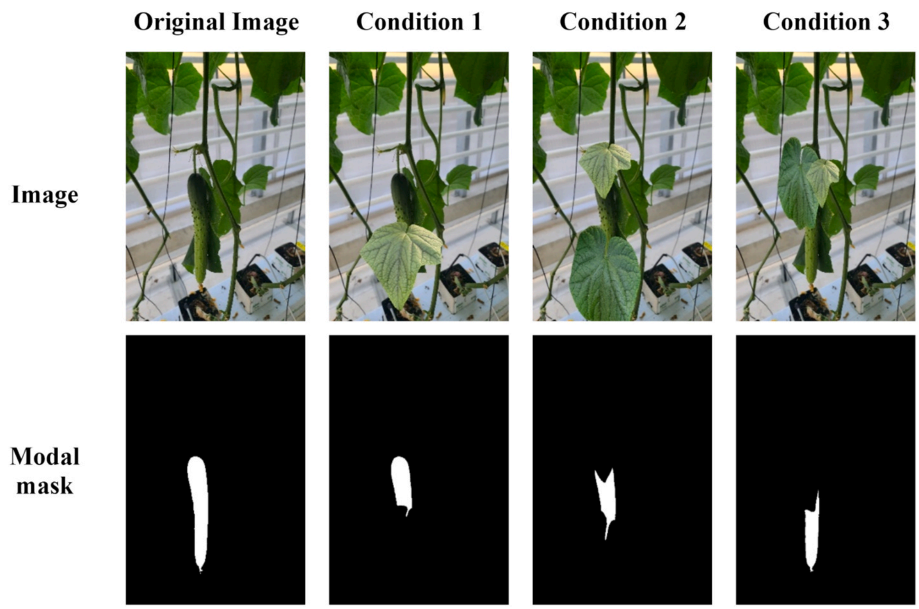


Fig. 2. Comparison of generated images and corresponding modal masks with the original image across different occlusion conditions.

parameters, leaf masks were systematically resized and positioned in different positions to simulate light, moderate, and dense occlusion scenarios.

Fig. 3 presents representative examples of occluded images for each experimental condition. In these visualizations, red bounding boxes indicate the amodal cucumber regions, green boxes denote visible regions, and black boxes represent synthesized leaves generated based on predefined occlusion parameters. The integration of leaf morphology characteristics, plant developmental stages, and practical agricultural insights enhance the dataset, providing a robust foundation for advancing amodal segmentation models.

Including diverse occlusion patterns enhances model generalization, enabling effective handling of real-world challenges such as dense foliage and complex occlusions commonly encountered in agricultural fields. Table 1 shows a summary of the synthesis process, which generated 18,018 instances—comprising 8,008 cucumbers and 10,010 leaves—culminating in the O2 dataset construction. This dataset significantly improves their robustness and applicability across diverse agricultural scenarios.

2.3.3. Brightness and contrast augmentation

Lighting variability is a significant factor in real-world agricultural settings, where illumination conditions fluctuate owing to changes in sunlight, shading, and weather patterns. To enhance model robustness under such conditions, Gamma Intensity Correction was applied during training (Ha et al., 2024). Images were randomly augmented to include one of three brightness conditions. In the dark condition, a gamma value of 0.3 was applied to simulate shaded or low-light environments. The original condition retained the natural illumination of the image with no gamma adjustment ($\gamma = 1.0$). For the bright condition, a gamma value of 3.0 was used to replicate high-exposure scenarios, such as intense sunlight or overexposure (Fig. 4).

The gamma values for dark and bright conditions are commonly used in low-light action recognition models, effectively simulating extreme lighting variations (Ha et al., 2024). While these values were selected for our experiments, the framework allows for adjustments based on specific lighting conditions or research objectives.

All occlusion conditions within the O2 dataset were maintained during brightness augmentation, and mask annotations remained unchanged to ensure consistent supervision. In addition to training on the combined dataset, separate models were trained exclusively on dark or bright image subsets.

2.4. Amodal instance segmentation model architecture

Amodal Instance Segmentation (AIS) addresses the challenge of detecting occluded cucumbers in natural environments, where significant portions of the target object are often obscured by leaves. AISFormer—a Transformer-based architecture proposed by Tran et al. (2022)—was adopted to segment visible and occluded regions of cucumbers. AISFormer is specifically designed to model complex interactions among various mask types—visible, occluder, amodal, and invisible—by leveraging region-of-interest (ROI) features extracted from input images.

2.4.1. Feature Encoding

Visual features were initially extracted from the synthetic cucumber-leaf dataset using a backbone network, such as ResNet. ROIAlign was then employed to isolate ROI features corresponding to the bounding boxes of detected cucumbers. These features were further refined through a deconvolution layer to enhance their resolution. The upsampled features were subsequently flattened and combined with positional encodings, serving as inputs to the Transformer encoder. The encoder effectively captured local and global dependencies within the ROI, generating enriched feature representations for accurate mask estimation.

2.4.2. Mask Transformer decoding

AISFormer employs learnable query embeddings to handle distinct mask types, including visible, occluder, and amodal. These queries are processed through a Transformer decoder, employing self-attention to model inter-query relationships and cross-attention to capture query-ROI interactions. This attention mechanism enables effective framework exploitation of contextual information, thereby enhancing mask prediction accuracy. The mask transformer decoding process is mathematically represented in Equation (1):

$$Q = \mathscr{D}_{\beta}(Q, F_e) = \mathscr{A}_{cross}\left(F_e, \left(\mathscr{A}_{self}(Q) \oplus Q\right)\right) \tag{1}$$

where the learnable query embeddings Q are updated through the self- $\mathscr{A}_{self}$ and cross-attention $\mathscr{A}_{cross}$ mechanisms. These mechanisms integrate information from the query and encoded features F_e, enabling effective contextual relationship modeling for accurate mask prediction.

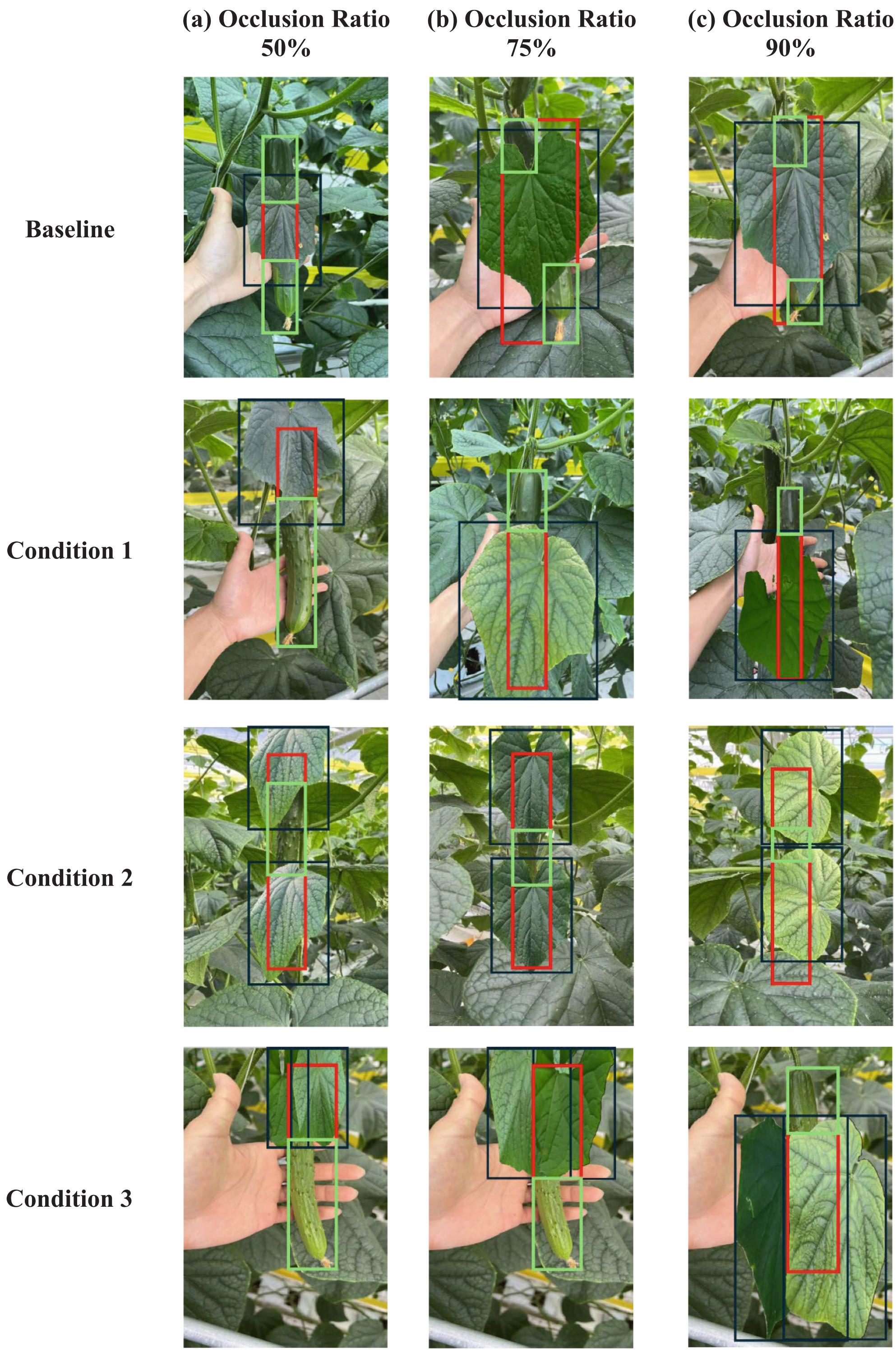


Fig. 3. Visualization of the O2 synthesized image dataset under various occlusion conditions. Red bounding boxes indicate amodal cucumber regions, green boxes indicate visible regions and black boxes highlight synthesized leaves generated based on the specified occlusion parameters.

2.4.3. Invisible mask embedding

Amodal masks are essential for capturing visible and occluded regions of cucumbers, such as those hidden by leaves. AISFormer incorporates an invisible mask embedding module that combines visible and amodal mask embeddings to infer the occluded regions. This integration enhances the prediction of occluded cucumber parts, which is a crucial amodal segmentation component in agricultural scenarios.

2.4.4. Mask prediction

In the final step, the framework generated four distinct masks: visible, occluder, amodal, and invisible. By combining the original ROI features with the Transformer-encoded outputs, pixel-wise ROI embeddings were produced, which were then processed to generate each mask type. This comprehensive segmentation approach enabled the framework to effectively handle complex occlusions, ensuring accurate AIS of cucumbers occluded by leaves. When AISFormer was integrated

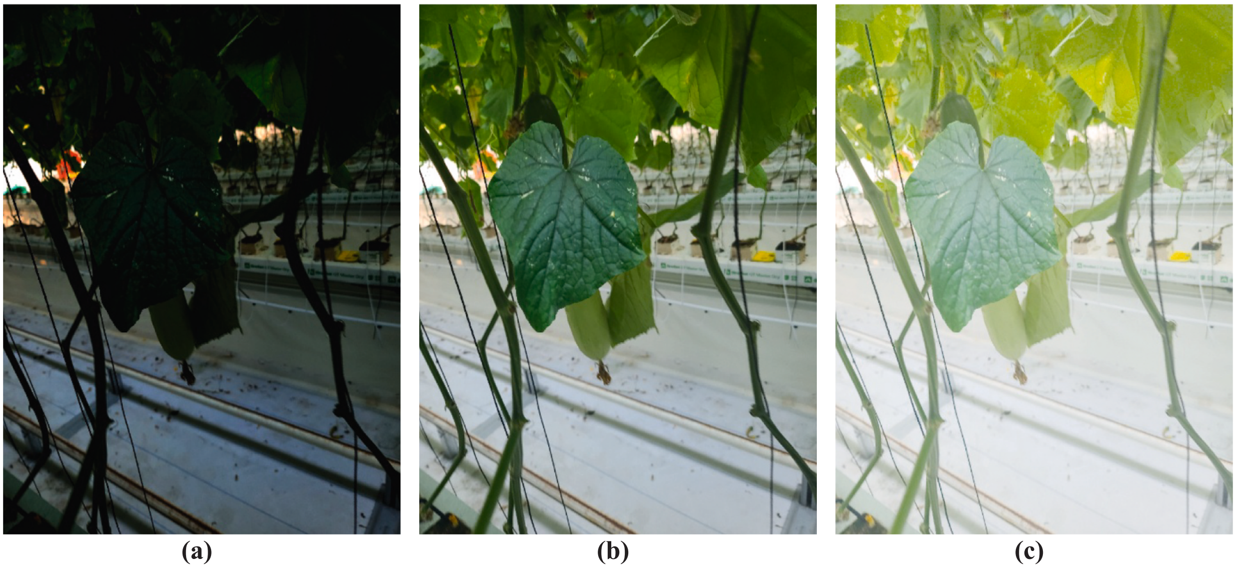


Fig. 4. Brightness and contrast augmentation applied to cucumber images. (a) Dark, (b) original, and (c) bright lighting conditions.

with the O2 dataset, it achieved robust performance in detecting and segmenting occluded cucumbers under dynamic occlusion conditions, demonstrating the real-world applicability of our approach in agricultural settings.

2.5. Evaluation metrics

We evaluated the proposed amodal segmentation framework using a set of established segmentation metrics to assess various performance aspects. These metrics include Intersection over Union (IoU), mean Average Precision (mAP), Average Precision (AP) at different occlusion levels, and Average Recall (AR). These metrics allowed us to measure the accuracy, robustness, and reliability of the segmentation results across diverse conditions.

2.5.1. Intersection over union

IoU measures the overlap between predicted and ground truth masks, serving as a straightforward metric for alignment accuracy (Equation (2)). IoU is calculated as the ratio of the intersection area to the union area of the predicted mask (M_{pred}) and the ground truth mask (M_{true}).

$$IoU = \frac{|M_{pred} \cap M_{true}|}{|M_{pred} \cup M_{true}|} \tag{2}$$

IoU is computed independently for visible, occluded, and amodal masks, enabling a detailed assessment of segmentation performance across various scenarios. This metric not only measures prediction accuracy but is also integral to mAP, AP, and AR calculations.

2.5.2. Mean average precision

mAP evaluates the precision-recall performance of the model across various IoU thresholds (Equation (3)). mAP is calculated as the mean of the AP values at specific IoU thresholds, typically ranging from 0.5 to 0.95 in increments of 0.05. This metric assesses the ability of the model to consistently predict accurate masks while minimizing false positives and false negatives. mAP is particularly useful for scenarios involving varying object scales and occlusions.

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{3}$$

2.5.3. Average precision by object scales

To provide a more detailed evaluation, AP was computed based on object scales to evaluate performance across different object sizes. Objects were categorized as small (AP_s, area $< 32 \times 32$ pixels), medium (AP_m, 32×32 pixels $\leq$ area $< 96 \times 96$ pixels), and large (AP_l, area $\geq 96 \times 96$ pixels). These metrics ($AP_s, AP_m, andAP_l$) offer valuable insights into the ability of the model to effectively handle objects of varying sizes.

2.5.4. Average Recall

For amodal segmentation, AR was calculated to evaluate the effectiveness of the model in detecting instances comprehensively across various occlusion scenarios. AR reflects the highest recall, averaged over several IoU thresholds. AR is computed for a set number of detections per image, such as AR10 and AR100, corresponding to a maximum of 10 and 100 detections, respectively. This metric assesses the ability of the model to retrieve occluded objects under realistic conditions.

3. Results and discussion

We conducted several experiments to assess the effectiveness of our amodal segmentation framework for cucumbers, particularly in the presence of partial occlusions caused by foliage. Section 3.1 shows the experimental setup, including dataset partitioning and hyperparameter configurations. Section 3.2 presents performance evaluations on the proposed O2 dataset, emphasizing amodal and modal segmentation metrics. Section 3.3 shows the generalization ability of the model under dynamic occlusion scenarios, wherein portions of the cucumbers were continuously obscured by leaves. Finally, Section 3.4 offers qualitative results, illustrating the robustness of the proposed method in addressing real-world occlusions.

3.1. Experiment setup

All experiments were conducted using the O2 dataset, comprising 768×1024 resolution images of cucumber plants annotated with visible (modal) and complete (amodal) masks. The training was performed with a batch size of 12 and an initial learning rate of 0.0025, regulated using a WarmupMultiStepLR scheduler (warmup = 1,000 iterations). Optimization was performed using the AdamW optimizer with a momentum parameter of 0.9. The loss function combined cross-entropy terms for classification and mask prediction tasks. The encoder was constructed using a Generalized RCNN backbone, modified to include an amodal segmentation head. The training was performed on a single NVIDIA RTX A6000 GPU, with the dataset partitioned into dedicated training and testing sets that represent varying occlusion levels. Performance was measured using mAP and mean AR at multiple IoU thresholds.

3.2. Performance comparison on our dataset (O2)

Table 3 presents the performance of the proposed model evaluated on the O2 dataset under three brightness conditions (Dark, Original, and Bright) and four occlusion scenarios (Baseline, Condition (1)–(3). Amodal and modal segmentation metrics—mAP, AP50, AP75, and mAR—were employed to assess the ability of the model to handle visible and occluded regions.

Across all lighting conditions, Condition (2) consistently demonstrated the most reliable amodal performance in the Original (mAP = 0.517, AP50 = 0.864), suggesting that when occlusions are moderately distributed at the top and bottom, the model can better infer the hidden regions. Under Bright conditions, Condition (3) exhibited strong amodal performance (mAP = 0.497), indicating that in some high-contrast scenarios, the model could still effectively manage complex overlapping occlusions.

For modal segmentation, which targets only visible regions, Conditions (1) and (3) under the Bright setting achieved the highest scores (e.g., mAP = 0.646 and 0.644, AP50 = 0.937 and 0.933, respectively), indicating that strong lighting can enhance the performance of the model when sufficient visible areas are present. Conversely, the lowest amodal mAP was observed in Condition (2) under Dark lighting (mAP = 0.328), highlighting the challenge of predicting occluded shapes when visibility and lighting conditions are poor.

These findings illustrate the combined effect of spatial occlusion and lighting variation. While Bright lighting can enhance modal segmentation, the performance of amodal prediction—particularly under dense occlusion (Condition (3))—varies with brightness.

To address these limitations, recent studies suggest incorporating spatial attention mechanisms and geometric priors, which help models focus on contextual features and enhance the inference of fully occluded objects (He et al., 2022; Li et al., 2022). The findings suggest that training with diverse lighting conditions, such as those achieved through Gamma augmentation, enhances the robustness of the segmentation model, enabling it to generalize more effectively across real-world agricultural lighting variations.

Fig. 5 illustrates the correlation between predicted and ground truth amodal areas, offering valuable insights into segmentation performance.

In the baseline scenario (Fig. 5a), with minimal occlusion, the predictions closely aligned with the ground truth, offering a reliable reference for evaluating segmentation accuracy. Under Condition (1) (Fig. 5b), where a single leaf partially occluded the top or bottom of the cucumber, the predictions remained highly accurate. This condition was easier for the model to handle, as reflected in the higher scores achieved. Condition (2) (Fig. 5c), where two separate leaves occluded the top and bottom of the cucumber, presented a slightly greater challenge. While predictions were less accurate compared to those in Condition (1), the model still performed reliably by leveraging visible areas to infer the occluded regions. Condition (3) (Fig. 5d) was the most complex, with overlapping leaves that significantly obscured the cucumber.

Table 4 complements these qualitative observations by presenting the mean amodal and modal scores across all occlusion scenarios. The results confirm that spatial agreement between the predicted and ground truth masks is well maintained in the baseline and Condition (1), but gradually deteriorates as occlusion becomes more severe in Conditions (2) and (3). This indicates that the decline in amodal IoU under heavy occlusion reflects an increasing difficulty in recovering the complete shape of the object.

Overall, these findings demonstrate the model's robustness under moderate occlusion while highlighting areas for improvement in more complex cases involving dense and overlapping foliage.

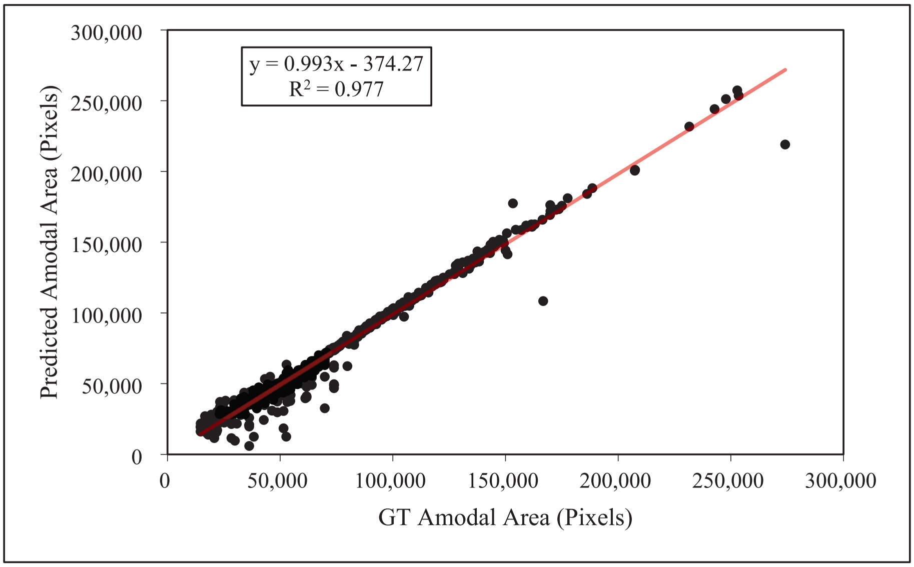
3.3. Performance comparison in dynamic occlusion conditions

While Section 3.2 highlights the generalization capability of a unified model across varying occlusion conditions, Section 3.3 reveals whether training models under condition-specific settings can further enhance performance.

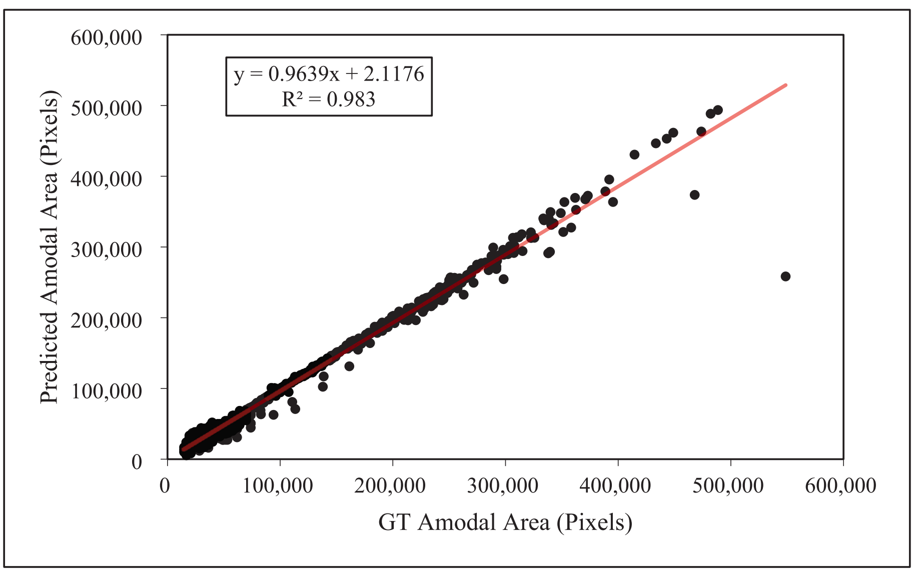
Table 3
Comparison of segmentation performance on the O2 dataset.

Brightness	Segmentation type	Occlusion condition	mAP	AP50	AP75	mAR
Dark	**Amodal**	**Baseline**	0.435	0.701	0.464	0.483
		Condition (1)	0.449	0.792	0.412	0.506
		Condition (2)	0.328	0.688	0.308	0.374
		Condition (3)	0.419	0.760	0.401	0.476
	Modal	**Baseline**	0.401	0.567	0.460	0.439
		Condition (1)	0.574	0.877	0.637	0.629
		Condition (2)	0.392	0.726	0.381	0.439
		Condition (3)	0.550	0.872	0.605	0.611
Original	**Amodal**	**Baseline**	0.443	0.781	0.430	0.536
		Condition (1)	0.493	0.771	0.497	0.548
		Condition (2)	0.517	0.864	0.483	0.577
		Condition (3)	0.383	0.733	0.380	0.430
	Modal	**Baseline**	0.537	0.834	0.586	0.630
		Condition (1)	0.449	0.637	0.491	0.504
		Condition (2)	0.676	0.951	0.782	0.727
		Condition (3)	0.480	0.800	0.505	0.527
Bright	**Amodal**	**Baseline**	0.461	0.730	0.501	0.511
		Condition (1)	0.497	0.856	0.491	0556
		Condition (2)	0.342	0.680	0.345	0.401
		Condition (3)	0.497	0.843	0.472	0.551
	Modal	**Baseline**	0.424	0.614	0.490	0.471
		Condition (1)	0.646	0.937	0.777	0.699
		Condition (2)	0.425	0.752	0.447	0.486
		Condition (3)	0.644	0.933	0.729	0.695

Abbreviations: mAP, mean Average Precision; mAR, mean Average Recall.



(a)



(b)

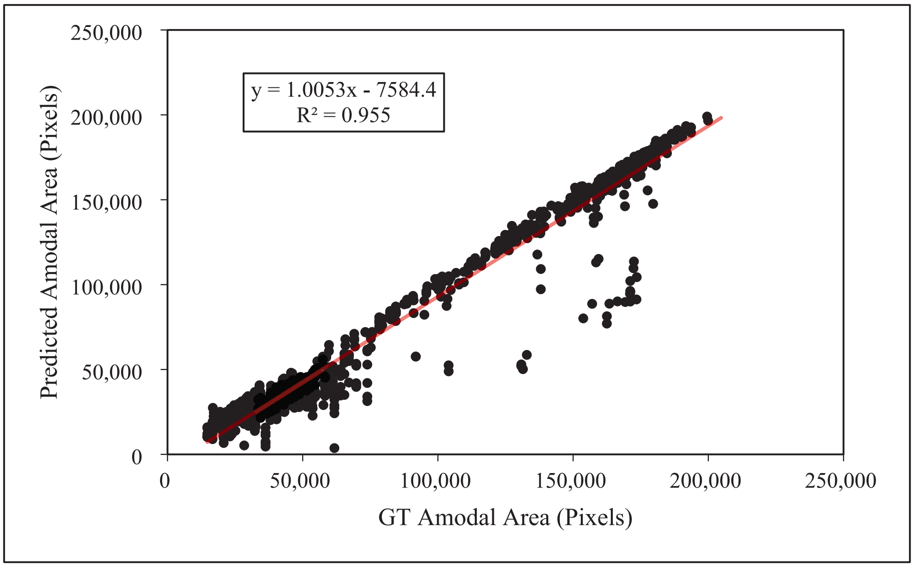
Fig. 5. Correlation between ground truth and predicted amodal segmentation. (a) Baseline, (b) Condition (1), (c) Condition (2), and (d) Condition (3).

Table 5 presents the performance of the model under dynamic occlusion scenarios, simulating real-world agricultural environments with shifting leaf positions. These experiments were designed to evaluate the robustness and adaptability of the model across varying occlusion complexities and lighting conditions. Each model was trained and tested under consistent, condition-specific settings to evaluate the effectiveness of targeted segmentation strategies.

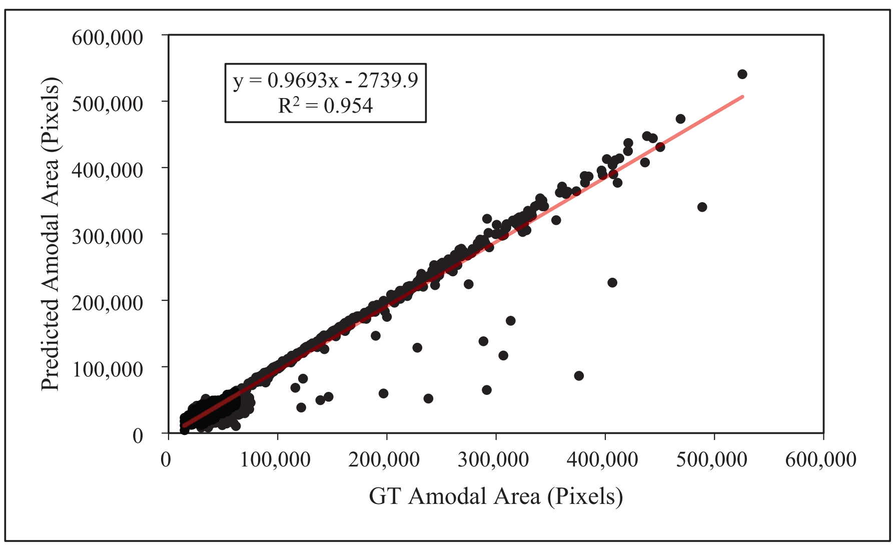
Under Original lighting, the baseline configuration for amodal segmentation achieved the highest performance (mAP = 0.581, AP50 = 0.906), confirming that minimal occlusion and balanced illumination offer optimal conditions for predicting the complete shape of cucumbers. However, segmentation performance declined in Conditions (2) and (3), where occlusions were spatially dispersed or overlapping. For instance, amodal mAP decreased to 0.540 and 0.548, respectively, under these conditions. This demonstrates the increased challenge of inferring occluded regions when object visibility is fragmented.

For modal segmentation, which focuses solely on visible regions, the best performance was also observed under original lighting, particularly in Condition (1) (mAP = 0.729, AP50 = 0.968) and Condition (3) (mAP = 0.722, AP50 = 0.968). This suggests that the model can effectively localize visible parts, especially when occlusions are concentrated in areas that do not obscure critical visual features.

Under Bright conditions, Condition (1) maintained strong modal segmentation (mAP = 0.662, AP50 = 0.941), while amodal performance remained moderate (mAP = 0.515). Conversely, Dark conditions led to an overall decline in accuracy. For example, in Condition (2), the amodal mAP dropped to 0.413 and the modal mAP to 0.515, highlighting the compounded challenge introduced by poor lighting and



(c)



(d)

Fig. 5. (*continued*).

Table 4
Mean Amodal and Modal IoU Scores Across Occlusion Conditions.

Condition	mIoU	
	Amodal	Modal
Baseline	0.939	0.924
Condition (1)	0.912	0.928
Condition (2)	0.877	0.892
Condition (3)	0.876	0.920

dispersed occlusion. These findings are consistent with those of previous research by Pegoraro and Pflugfelder (2020), which emphasized that dispersed occlusions significantly degrade segmentation accuracy owing to disrupted spatial continuity. When occlusions occur at multiple, non-contiguous points—as seen in Condition (2)—the model struggles to integrate fragmented visual cues, limiting its ability to reconstruct complete object shapes. Conversely, overlapping occlusions, such as those in Condition (3), tend to create more predictable shadowing patterns, which the model can interpret more effectively.

Overall, the results in Table 5 illustrate that segmentation performance is highly sensitive to the occlusion distribution and lighting conditions. While condition-specific training enhances adaptability, scattered occlusions and poor lighting remain significant challenges. This suggests the need for future research to incorporate temporal consistency, geometric priors, or attention mechanisms, enabling the model to more effectively reason through fragmented observations and enhance performance in real-world scenarios.

3.4. Qualitative results in dynamic occlusion conditions

Qualitative assessments (Fig. 6) further validate the quantitative findings. Our model consistently predicted the hidden edges of

Table 5
Segmentation performance under dynamic occlusion conditions.

Brightness	Segmentation type	Occlusion condition	mAP	AP50	AP75	mAR
Dark	Amodal	Baseline	0.484	0.781	0.491	0.547
		Condition (1)	0.505	0.862	0.462	0.698
		Condition (2)	0.413	0.797	0.384	0.445
		Condition (3)	0.479	0.846	0.452	0.535
	Modal	Baseline	0.437	0.666	0.469	0.497
		Condition (1)	0.649	0.929	0.747	0.735
		Condition (2)	0.515	0.842	0.560	0.543
		Condition (3)	0.640	0.923	0.748	0.668
Original	Amodal	Baseline	0.581	0.906	0.583	0.630
		Condition (1)	0.564	0.887	0.529	0.606
		Condition (2)	0.540	0.938	0.496	0.587
		Condition (3)	0.548	0.881	0.521	0.590
	Modal	Baseline	0.532	0.861	0.505	0.580
		Condition (1)	0.729	0.968	0.840	0.760
		Condition (2)	0.626	0.943	0.727	0.673
		Condition (3)	0.722	0.968	0.853	0.750
Bright	Amodal	Baseline	0.510	0.806	0.523	0.565
		Condition (1)	0.515	0.872	0.484	0.575
		Condition (2)	0.432	0.824	0.399	0.462
		Condition (3)	0.502	0.866	0.473	0.555
	Modal	Baseline	0.457	0.692	0.492	0.510
		Condition (1)	0.662	0.941	0.768	0.715
		Condition (2)	0.543	0.864	0.606	0.564
		Condition (3)	0.665	0.940	0.761	0.706

Abbreviations: mAP, mean Average Precision; mAR, mean Average Recall.

cucumbers, even when significant portions were partially occluded. These findings highlight the practical effectiveness of our method in real-world agricultural settings.

However, it is crucial to recognize the limitations of the current approach. While the model performs well in most conditions, handling dense and scattered occlusions remains a challenge, especially when the occluded regions are fragmented or poorly illuminated. For example, in Condition (2) under low-light conditions, performance decreased due to scattered occlusions disrupting spatial continuity.

Additionally, performance of model is influenced by the distribution of occlusions. Overlapping occlusions (Condition (3)) tend to create more predictable patterns, while scattered occlusions (Condition (2)) make it more difficult to reconstruct the full shape of the cucumber. This issue is amplified under varying lighting conditions, as changes in illumination, such as low-light or bright settings, affect segmentation accuracy, as shown in Table 5. These results highlight the need for further improvement in the model's ability to handle dynamic conditions.

In summary, the model works well in most agricultural settings, but future work should explore incorporating spatial attention mechanisms or temporal consistency to better address challenges related to complex occlusions and lighting variations.

3.5. Ablation study

To evaluate the general applicability of the proposed framework across different model capacities, we trained a series of YOLOv5 segmentation models using the proposed dataset. YOLOv5 was selected due to its proven performance in real-time agricultural tasks, such as pest detection (Qi et al., 2024) and food segmentation (Phadke et al., 2024), as well as its scalable architecture, which ranges from nano (YOLOv5n) to extra-large (YOLOv5x) (Zaidi et al., 2022). This ablation study aimed to assess whether the proposed framework could generate training data suitable for a wide range of segmentation model complexities.

Table 6 presents the segmentation performance of each YOLOv5 model, along with AISFormer, evaluated using both modal and amodal masks. All models were trained and validated on the entire dataset. As model size increased from YOLOv5n to YOLOv5x, both mAP and AP50 improved consistently, with YOLOv5x achieving the highest performance among the YOLO variants (mAP = 0.489, AP50 = 0.741). AISFormer slightly outperformed YOLOv5x (mAP = 0.490, AP50 = 0.808), indicating the potential of transformer-based architectures for capturing both visible and occluded regions. These results confirm that the proposed framework generates high-quality training data that generalizes well across diverse model capacities and segmentation tasks.

3.6. Study limitations

Despite the promising results, this study has several important limitations that should be acknowledged.

Environmental sensitivity: The modal showed notable performance variations under different conditions. Specifically, accuracy devreased significantly in dark lighting environments and scattered occlusion scenarios (Condition 2). This sensitivity suggests that the current approach requires further optimization before deployment in challenging field conditions where lighting and occlusion patterns can be unpredictable.

Limited scope and generalizability: The evaluation focused primarily on cucumber crops within controlled experimental settings. While the framework achieved high performance with occlusion rates up to 90%, the findings may not readily transfer to other crop types with different shapes, growth patterns, or morphological characteristics. The framework's effectiveness across diverse agricultural crops remains to be validated.

Validation limitations: Long-term field validation across multiple growing seasons and diverse environmental conditions has not been conducted. Without such extensive testing, the framework's consistency and reliability for practical agricultural deployment cannot be fully established.

These limitations highlight important directions for future research to enhance the framework's real-world applicability and ensure robust performance across diverse agricultural settings.

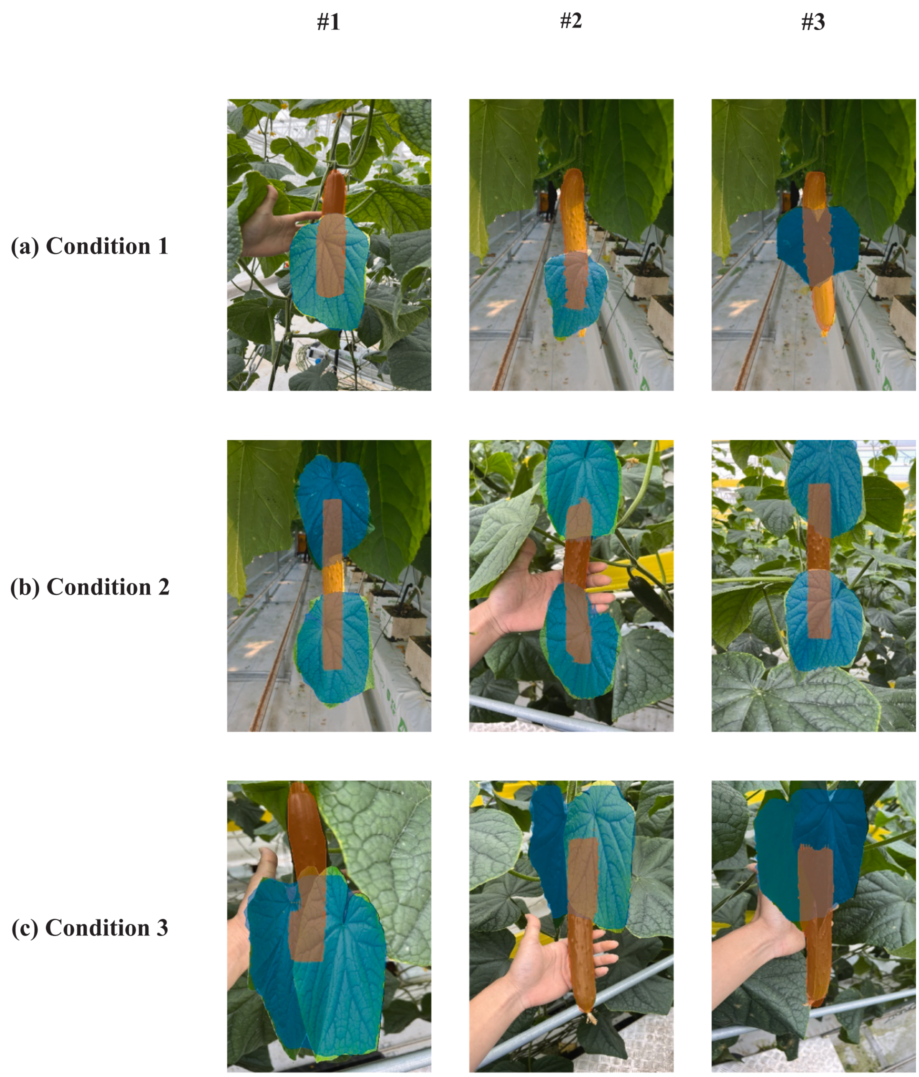


Fig. 6. Qualitative comparison of segmentation predictions on the proposed dataset and real-world cucumber images. Rows represent Conditions (a) 1, (b) 2, and (c) 3. Blue masks indicate predicted synthesis leaves, while orange masks indicate predicted cucumber regions.

Table 6
Segmentation performance of YOLOv5 models and AISFormer on the automatically generated dataset.

Model	mAP	AP50
YOLOv5n	0.321	0.669
YOLOv5s	0.387	0.678
YOLOv5m	0.423	0.695
YOLOv5l	0.466	0.730
YOLOv5x	0.489	0.741
AISFormer	0.490	0.808

4. Conclusion

This study presents a novel framework for generating automatic synthetic occlusion datasets and performing amodal segmentation, offering a robust solution to the persistent challenges of occlusion and dense foliage in agricultural environments. Our approach addresses the limitations of conventional segmentation methods, which often underperform in complex real-world agricultural conditions, by simulating realistic occlusion scenarios that vary in leaf size, position, and overlap. Additionally, incorporating Gamma-based brightness augmentation enhances model robustness by simulating dynamic lighting conditions, including dark and bright environments, thereby capturing natural variability present in outdoor agricultural settings.

The proposed framework demonstrates strong multi-class segmentation capabilities and accurately reconstructs occluded crop shapes, demonstrating high efficacy for complex crops such as cucumbers. By systematically introducing single, double, and overlapping leaf occlusion scenarios, the framework provides a reliable foundation for evaluating and enhancing segmentation models across varying levels of occlusion complexity. This structure enhances their reliability and robustness. The model performed best under bright lighting conditions, particularly in scenarios involving overlapping occlusions, whereas performance in dark settings posed greater challenges, highlighting the sensitivity of the model to occlusion and lighting conditions.

While this study primarily focuses on cucumber segmentation, the proposed framework is designed with extensibility in mind, offering the potential for future adaptation to other crop types and agricultural environments. Through continued refinement within the cucumber domain and systematic experimentation across various crops, the framework can be further validated and optimized for broader agricultural applications.

To integrate the proposed model into existing agricultural automation pipelines, future research could focus on deploying it in robotic systems for crop detection, harvest planning, and yield estimation. By optimizing the framework for real-time processing and integrating it with monitoring systems in automated harvesting robots, the model can facilitate efficient and scalable precision agriculture operations. This would enable the practical application of the segmentation model in tasks such as automatic crop classification, fruit picking, and yield forecasting.

Future research could explore incorporating dynamic environmental factors, such as changing weather conditions, crop growth stages, and temporal changes, into the framework. These advancements would further enhance its versatility and establish amodal segmentation as a crucial tool for addressing agricultural challenges, driving innovation, and fostering sustainable farming practices.

CRediT authorship contribution statement

Jin-Ho Son: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Hojun Song:** Project administration, Methodology. **Chae-yeong Song:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Minse Ha:** Validation, Formal analysis, Software. **Dabin Kang:** Visualization, Methodology. **Yu-Shin Ha:** Writing – review & editing, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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